[INCH-POUND]

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11 February 1991
SUPERSEDING
MIL-R-39032C
25 January 1988

MILITARY SPECIFICATION

RESISTORS, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the preservation, packing, and container marking of all types of resistors and associated accessories which are primarily FSC 5905.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin.
NN-P-71	-	Pallets, Material Handling, Wood Stringer Construction, 2-Way and 4-Way(Partial).
QQ-S-781	-	Strapping, Steel, and Seals.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-R-11	-	Resistors, Fixed, Composition (Insulated) General Specification for.
MIL-R-26	-	Resistors, Fixed, Wire-wound (Power Type), General Specification for.
MIL-R-93	-	Resistors, Fixed, Wire-wound (Accurate), General Specification for.
MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Sleeves and Tubing.
MIL-R-122	-	Resistors, Fixed, Precision, Established Reliability, General Specification for.
MIL-R-874	-	Resistor Networks, Fixed, Film, Established Reliability, General Specification for.
MIL-R-914	-	Resistor Networks, Fixed, Film, Surface Mount, Established Reliability, General Specification for.
MIL-D-3464	-	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-R-10509	-	Resistors, Fixed, Film (High Stability), General Specification for.
MIL-R-18546	-	Resistors, Fixed, Wirewound (Power Type, Chassis Mounted), General Specification for.
MIL-R-22097	-	Resistors, Variable, Nonwire-Wound (Adjustment Type), General Specification for.
MIL-R-22684	-	Resistors, Fixed, Film, Insulated, General Specification for.
MIL-T-23648	-	Thermistor, (Thermally Sensitive Resistor) Insulated General Specification for.
MIL-R-27208	-	Resistor, Variable, Wire-wound (Adjustment Type), Lead-Screw Actuated.
MIL-R-39005	-	Resistors, Fixed, Wire-wound (Accurate), Established Reliability, General Specification for.
MIL-R-39007	-	Resistors, Fixed, Wire-wound (Power Type), Established Reliability, General Specification for.
MIL-R-39008	-	Resistors, Fixed, Composition (Insulated), Established Reliability General Specification For.
MIL-R-39009	-	Resistors, Fixed, Wire wound (Power Type, Chassis Mounted), Established Reliability, General Specification for.
MIL-R-39015	-	Resistors, Variable, Wire-wound (Lead Screw Actuated), Established Reliability General Specification For
MIL-R-39017	-	Resistors, Fixed, Film (Insulated) Established Reliability, General Specification for.
MIL-R-39035	-	Resistors, Variable, Nonwire-wound (Adjustment Type), Established Reliability, General Specification for.
MIL-R-49462	-	Resistors, Fixed, Film, High Voltage, General Specification for.
MIL-R-49465	-	Resistors, Fixed, Metal Element (Power Type), (Very Low Resistance Values), General Specification for.
MIL-R-55182	-	Resistors, Fixed, Film, Established Reliability General Specification for.
MIL-R-55342	-	Resistors, Fixed, Film, Chip, Established Reliability, General Specification for.
MIL-B-81705	-	Barrier Materials, Flexible, Electrostatic Protective, Heat Sealable.
MIL-R-83401	-	Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistors, General Specification for.
MIL-R-87254	-	Resistor, Fixed, Film, High Reliability General Specification for.

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STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1189 - Standard Department of Defense Bar Code Symbology.
MIL-STD-1686 - Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
MIL-STD-2073-1 - DOD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-2073-2 - Packaging Requirement Codes
MS-20003 - Indicator, Humidity, Card, Three Spot, Impregnated Areas (Cobaltous Chloride).
MIL-STD-45662 - Calibration Systems Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Shipments to Government activities.

3.1.1 General. The packaging terms used herein shall be in accordance with the definitions listed in MIL-STD-129 and MIL-STD-2073-1. The following general requirements apply, as applicable, to levels A, B, and C.

3.1.1.1 Pairs and sets. Resistors furnished in pairs or sets under one national stock number (NSN) shall be unit packed as one pair or one set, as applicable. Unless otherwise specified, unit of issue (e.g., each, pair, set, etc.) shall be individually unit packed.

3.1.1.2 Hardware. Hardware accompanying resistors shall be protected and enclosed within the unit packs in a manner that will not damage the resistors or packaging materials. When practical or when the resistors are not otherwise protected, the hardware should be mounted on each resistor.

3.1.1.3 Physical protection. Resistors and accessories shall be packaged in a manner that will ensure compliance with the applicable requirements of MIL-P-116 as well as those specified herein.

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3.1.1.4 Wrapping and cushioning of leads and terminals for non-ESD resistors. For other than axial-leaded resistors, the leads, terminals, and resistor bonds shall be protected by container design, die cut inserts, vials, or suitable noncorrosive supporting materials or devices. Leads or terminals shall extend outward and be maintained in a configuration as manufactured without causing undue loads or stresses capable of causing damage to the resistors. Materials used to maintain item position and lead configuration shall permit item removal and replacement without bending the leads. Materials used shall be noncorrosive materials and shall not crumble, flake, powder, or shed. These materials shall be in accordance with MIL-STD-2073-2, table IV for wrapping materials and table V for cushioning materials. Wrapping and cushioning for ESD resistors shall be in accordance with 3.1.8 herein. No special physical protection or lead protection is required for axial-leaded resistors.

3.1.1.5 Exterior containers. Exterior containers shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered or otherwise designated items to the greatest extent practicable.

3.1.2 Preservation. Preservation shall be in accordance with level A, B, or C as specified (see 6.2).

3.1.2.1 Level A.

3.1.2.1.1 Cleaning. Resistors and accessories shall be clean or be cleaned in accordance with MIL-P-116, process C-1 if necessary.

3.1.2.1.2 Drying. Resistors and accessories shall be dry or be dried in accordance with MIL-P-116 if necessary.

3.1.2.1.3 Preservatives. Contact preservatives shall not be used.

3.1.2.1.4 Unit packs.

3.1.2.1.4.1 Hermetically sealed resistors except established reliability types. Hermetically sealed resistors (including ballast tubes) shall be individually unit packed or as specified in the contract or order in accordance with MIL-P-116, method III. When the net weight of resistors exceeds 5 pounds (2.268 kilograms), the unit container shall conform to PPP-B-636 class weather-resistant, style optional, special requirements.

3.1.2.1.4.2 Nonhermetically sealed resistors and all established reliability types. These resistors shall be individually unit packed in accordance with the procedures specified in table I. Unless otherwise specified in the contract, resistors conforming to MIL-R-39017 shall be unit packed 5 each, those conforming to MIL-R-11 and MIL-R-39008, 10 each (see 6.2).

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Table I. Unit pack requirements for resistors intended for shipment to Government facilities.

Type	Physical protection	Unit pack	Bag/material construction	Marking	Additional requirements
ESD Sensitive Resistors					
Film chips MIL-R-55342	Pouch made from MIL-B-81705, type II material	1	Heat sealed MIL-B-117 bag, type I, class F, style I. Made from MIL-B-81705 material.	See 3.1.5	See 3.1.2.1.6 See 3.1.3
Film networks MIL-R-874 MIL-R-914 MIL-R-83401	MIL-P-116 method IA-8, formed support with surface resistivity $<10^{12}$ ohms per square	 1 1 1	See film chips	See 3.1.5	See 3.1.2.1.6 See 3.1.3
Ultra precision, axial leads MIL-R-122 MIL-R-55182 MIL-R-87254	No requirement	 1 1 1	See film chips	See 3.1.5	See 3.1.2.1.6 See 3.1.3
Non-ESD Sensitive Resistors					
Axial lead, film composition MIL-R-10509 MIL-R-22684 MIL-R-39017 MIL-R-49462	No requirement	 1 1 5 1	Heat sealed MIL-B-117, type I, class E or F, style 1, 2, or 3; type II, class E, style 1 or 3; type III, class E, style 1	See 3.1.5	See 3.1.2.1.6 See 3.1.3
Axial lead, composition MIL-R-11 MIL-R-39008	No requirement	 10 10	Same as axial lead, film	See 3.1.5	See 3.1.2.1.6 See 3.1.3

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Table I. Unit pack requirements for resistors intended for shipment to Government facilities - Continued.

Type	Physical protection	Unit pack	Bag/material construction	Marking	Additional requirements
Non-ESD Sensitive Resistors - Continued					
Axial lead, fixed and variable (including lug types)	No requirement		Same as axial lead, film	See 3.1.5	See 3.1.2.1.6 See 3.1.3
MIL-R-26		1			
MIL-R-93		1			
MIL-R-18546		1			
MIL-R-39005		1			
MIL-R-39007		1			
MIL-R-39009		1			
MIL-R-49465		1			
Temperature sensitive	No requirement		Same as axial lead, film	See 3.1.5	See 3.1.2.1.6 See 3.1.3
MIL-T-23648		1			
Variable, trim pots, excluding those with antifriction bearings	No requirement		Same as axial lead, film	See 3.1.5	See 3.1.2.1.6 See 3.1.3
MIL-R-22097		1			
MIL-R-27208		1			
MIL-R-39015		1			
MIL-R-39035		1			
Variable with antifriction bearings	MIL-P-116 method IIC, cushioning in accordance with 3.1.1.4, MIL-D-3464 desiccant, MS-20003 humidity indicator	1	Same as axial lead, film	See 3.1.5	See 3.1.2.1.6 See 3.1.3

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3.1.2.1.5 Accessories (when separately acquired). Accessories, such as mounting brackets, shall be unit packed one each in accordance with MIL-P-116, method III. Unit pack quantities for other than one each shall be as specified (see 6.2). Cushioning is not required for accessories.

3.1.2.1.6 Supplementary containers. Resistors and accessories unit packed in a bag or envelope exceeding 15 cubic inches (245.85 cubic centimeters) shall be placed within a supplementary container conforming to variety 2 of PPP-B-566 or PPP-B-676.

3.1.2.1.7 Intermediate packs. When the unit pack is a bag of any type or the unit pack is less than 64 cubic inches (1,049 cubic centimeters), uniform quantities of resistors or accessories of the same NSN shall be placed in intermediate containers conforming to variety 2 of PPP-B-566, or variety 2 of PPP-B-676; or PPP-B-636, class weather resistant. Alternatively, nonweather or nonwater resistant versions of these containers may be used provided they are overwrapped with waterproof barrier materials. Intermediate containers shall contain multiples of five unit packs not to exceed 100 unit packs. No intermediate packs are required when the total quantity shipped to a single destination will result in only one intermediate pack per shipping container.

3.1.2.2 Level B. The level B preservation for resistors and accessories shall be identical to that specified for level A except that submethod IC-1 or IC-3 shall be used to meet the unit pack requirements for all resistors not specified in table I. In addition, any variety or class of the unit, supplementary, and intermediate containers specified for level A may be used.

3.1.2.3 Level C. In addition to the applicable provisions of 3.1.1, the level C preservation for resistors and accessories shall conform to the MIL-STD-2073-1 requirements for this level.

3.1.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.1.3.1 Level A. Resistors and accessories, preserved as specified in 3.1.2 shall be packed in wood containers conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. The requirements for level B packing shall be used when the total quantity of a stock numbered resistor or accessory for a single destination does not exceed a packed volume of 1 cubic foot (0.02832 cubic meter).

3.1.3.2 Level B. Resistors and accessories, preserved as specified in 3.1.2, shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. The requirements for box closure, waterproofing, and reinforcing shall be in accordance with method V of the PPP-B-636 appendix.

3.1.3.3 Level C. Resistors and accessories, preserved as specified in 3.1.2, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the PPP-B-636 appendix.

3.1.4 Unitized loads. Unless otherwise specified (see 6.2), unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet (1.1328 cubic meters) or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

3.1.4.1 Level A. Resistors, packed as specified in 3.1.3.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a wood cap (storage aid 5) positioned over each load.

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3.1.4.2 Level B. Resistors, packed as specified in 3.1.3.2, shall be unitized as specified in 3.1.4.1 except that weather resistant fiberboard caps (storage aid 4) shall be used in lieu of wood caps.

3.1.4.3 Level C. Resistors, packed as specified in 3.1.3.3, shall be unitized as specified in 3.1.4.2 except that the fiberboard caps shall be class domestic.

3.1.5 Marking. The following marking is mandatory for shipments both to United States Government and non-Government activities.

3.1.5.1 Standard marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit, supplementary, intermediate, and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or Part or Identifying Number (PIN), as applicable, including the Commercial and Government Entity (CAGE) code, shall be marked on all unit and intermediate packs in accordance with the identification marking provisions of MIL-STD-129. When specified in the contract (see 6.2), the marking of domestic shipments for civil agencies shall be in accordance with FED-STD-123.

3.1.5.2 Special marking. In addition to the marking requirements of 3.1.5.1, and regardless of the level or type of packaging specified, all of the unit, supplementary, intermediate, and exterior containers and unitized loads for those resistors specified in 3.1.8 shall be marked with the sensitive electronic device unit pack label and the sensitive electronic device caution label (as applicable) specified in MIL-STD-129.

3.1.5.3 Additional unit pack marking of FSC 5905 commodity specification resistors. Unit packs of resistors conforming to FSC 5905 commodity specifications shall additionally be marked with the following:

a. Date code and (when authorized) JAN marking.

b. Manufacturer's production lot code.

3.1.5.4 Small item additional unit pack marking. MIL-R-55342 resistor unit packs shall contain the warning statement "Use caution when opening, item is very small and may require magnification to be seen." The warning statement shall be applied by label or by marking so as to be readily visible before opening the bag.

3.1.6 First article and quality conformance inspections. First article and quality conformance inspections and tests shall be required as specified in 4.4 and 4.5, respectively. Samples for these test shall be furnished in accordance with the procedures outlined in 4.4 and 4.5. The performance of the visual and dimensional inspections, rough handling tests, and leakage tests shall conform to the inspections and test outlined in 4.6.1, 4.6.2.1, and 4.6.2.2, respectively.

3.1.6.1 Functional requirements.

3.1.6.1.1 Rough handling test (when specified, see 6.2). When packs have been tested in accordance with 4.6.2.1, all materials and components comprising each pack shall be free from damage or evidence of misplacement which might affect the utility of the preservation method or pack. The resistors and associated accessories within the tested packs shall show no visible signs of damage. When specified in the contract (see 6.2), functional tests in accordance with the group A inspection requirements of the applicable commodity specification shall be conducted on those resistors subjected to the rough handling test to determine freedom from operational malfunction. The examination of the resistors tested under this group A inspection shall be in accordance with the visual and mechanical inspection requirements specified in the applicable resistor commodity specification.

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3.1.6.1.2 Leakage test (when applicable). When a barrier enclosed unit pack has been tested in accordance with 4.6.2.2, there shall be no evidence of moisture within the unit pack.

3.1.6.1.3 Workmanship. The quality of workmanship shall assure acceptance of the completed preservation, packing, and marking requirements in accordance with the inspections specified in section 4.

3.1.7 Army acquisitions (additional requirements for levels A, B, and C unit packs). The quantity per unit pack (QUP) shall be one, except as specified in 3.1.2.1.4. Unit and supplementary containers conforming to PPP-B-636, class weather-resistant shall be utilized in lieu of PPP-B-566 or PPP-B-676 containers.

3.1.7.1 Levels A, B, and C intermediate packs. Units packed in containers less than 64 cubic inches and bearing the same NSN, shall be placed in intermediate containers conforming to PPP-B-636, class weather-resistant. Level C intermediate containers shall conform to PPP-B-636, class domestic. Intermediate containers shall not exceed 40 pounds (18.144 kilograms) net weight, or a maximum size of 1.5 cubic feet (0.04248 cubic meter) with at least two dimensions not exceeding 16 inches (40.64 centimeters). Intermediate containers shall not be required when the total quantity to be shipped will result in only one intermediate pack per shipping container.

3.1.7.2 Levels A, B, and C packing. For wood containers conforming to PPP-B-601 or PPP-B-621, when the gross weight exceed 200 pounds (90.72 kilograms), or the container length and width is 48 x 24 inches (121.92 x 60.96 centimeters) or more and the weight exceed 100 pounds (45.36 kilograms), 3 x 4 inches (7.62 x 10.16 centimeters) skids (laid flat) shall be applied in accordance with the requirements of the container specification. Palletization shall be required when containers specified in 3.1.4 do not require skids; quantities per destination exceed either a total of 250 pounds or 113.4 kilograms (excluding the pallet) or a volume of 20 cubic feet (0.05563 cubic meter); and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified, except that container strapping may be omitted, shall be placed on a pallet, load type I conforming to MIL-STD-147. For levels B and C, unit or intermediate containers which meet these requirements may be palletized without further packing. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

3.1.7.3 Bar code marking. Bar code markings are required and shall be applied in accordance with MIL-STD-129 and MIL-STD-1189.

3.1.8 Electrostatic discharge (ESD) protection for resistors. A list of the military specifications for resistors that are sensitive to ESD is shown in table II. ESD sensitivity classifications, definitions, and ESD control program requirements are defined in MIL-STD-1686.

3.1.8.1 ESD protection for level of preservation A, B, and C. Resistors that are susceptible to electrostatic field forces shall be placed within barrier material conforming to MIL-B-81705, type I and heat sealed.

3.1.8.1.1 Bag material. Bags constructed to comply with ESD requirements shall be in accordance with MIL-B-117, type I, class A, style 2 or class F, style 1. In addition, MIL-B-81705 type III material is acceptable.

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3.2 Shipment to non-Government activities.

3.2.1 Non-ESD sensitive resistors. Shipments to original equipment manufacturers (OEM) and packaging distributors shall be in accordance with good commercial practices (see table III).

3.2.2 ESD sensitive resistors. Shipment of ESD sensitive resistors to OEM's and packaging distributors shall conform to the following requirements as shown in table III.

3.2.2.1 Packaging material surface resistivity. Tape and reel, waffle pack, and other type carriers used in packaging shall have a surface resistivity of less than 10^{12} ohms per square.

3.2.2.2 Container. Tape and reel, waffle pack, and other type carriers used in packaging, if required, shall be placed in a bag conforming to MIL-B-117, type I, class A, style 2 or class F, style 1 using barrier material conforming to MIL-B-81705 type I, II, or III as specified (see 6.2.2).

TABLE II. Specifications, styles, and PIN for ESD sensitive resistors.

Specification	Style designator	PIN
MIL-R-874/01	RZR10	M87401********
" /02	RZR20	M87402XXXXXXXX
" /03	RZR30	M87403XXXXXXXX
" /04	RZR40	M87404XXXXXXXX
" /05	RZR50	M87405XXXXXXXX
" /06	RZR60	M87406XXXXXXXX
" /07	RZR70	M87407XXXXXXXX
" /08	RZR80	M87408XXXXXXXX
" /09	RZR90	M87409XXXXXXXX
" /10	RZR95	M87410XXXXXXXX
MIL-R-87254/01	RSC50	M8725401XXXXXXX
" /02	RSC55	M8725402XXXXXXX
" /03	RSC60	M8725403XXXXXXX
" /04	RSC65	M8725404XXXXXXX
" /05	RSC90	M8725405XXXXXXX
MIL-R-122/01	RFP01	M122*01X*******
" /03	RFP03	M122X03XXXXXXXX
" /06	RFP06	M122X06XXXXXXXX
MIL-R-55342/01	RM0502	M55342*01******
" /02	RM0505	M55342*02******
" /03	RM1005	M55342*03******
" /04	RM1505	M55342*04******
" /05	RM2208	M55342*05*XXXXX
" /06	RM0705	M55342*06*XXXXX
" /07	RM1206	M55342*07*XXXXX
" /08	RM2010	M55342*08*XXXXX
" /09	RM2512	M55342*09*XXXXX
" /10	RM1010	M55342*10*XXXXX

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TABLE II. Specifications, styles, and PIN for ESD sensitive resistors - Continued.

Specification	Style designator	PIN
MIL-R-83401/01	RZ010	M8340101*******
" /02	RZ020	M8340102XXXXXXX
" /03	RZ030	M8340103XXXXXXX
" /04	RZ040	M8340104XXXXXXX
" /05	RZ050	M8340105XXXXXXX
" /06	RZ060	M8340106XXXXXXX
" /07	RZ070	M8340107XXXXXXX
" /08	RZ080	M8340108XXXXXXX
" /09	RZ090	M8340109XXXXXXX
" /10	RZ100	M8340110XXXXXXX
" /11	RZ110	M8340111XXXXXXX
" /12	RZ120	M8340112XXXXXXX
" /18	RZ150	M8340118XXXXXXX
" /19	RZ190	M8340119XXXXXXX
MIL-R-55182/01	RNC55	RNR55XXXXXXX
" /01	RNN55	RNN55XXXXXXX
" /01	RNR55	RNR55XXXXXXX
" /02	RNC57	RNC57XXXXXXX
" /02	RNN57	RNN57XXXXXXX
" /02	RNR57	RNR57XXXXXXX
" /03	RNC60	RNC60XXXXXXX
" /03	RNN60	RNN60XXXXXXX
" /03	RNR60	RNR60XXXXXXX
" /05	RNC65	RNC65XXXXXXX
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Table III. Shipments to non-Government facilities.

Type	Physical protection	Bag/material construction	Marking
ESD Sensitive Resistors			
Film chips MIL-R-55342 (tape and reel)	Use materials with surface resistivity less than 10^{12} ohms per square.	Tape and reel wrapped in MIL-B-81705 type I, II, or III material, if required (see 6.2.2).	See 3.1.5
(waffle pack)	Use carrier pack and top material with surface resistivity less than 10^{12} ohms per square.	Waffle pack made of or wrapped in MIL-B-81705 type I, II, or III material, if required (see 6.2.2).	See 3.1.5
Film networks MIL-R-874 MIL-R-914 MIL-R-83401 (tube pack)	Same as film chip	Tube packs made of or wrapped in MIL-B-81705 type I, II, or III material, if required (see 6.2.2).	See 3.1.5
Ultra precision, axial lead MIL-R-122 MIL-R-55182 MIL-R-87254 (tape and reel)	No requirement	Box or bag wrapped in MIL-B-81705 type I, II, or III material, if required (see 6.2.2).	See 3.1.5
(bulk)	No requirement	Box or bag wrapped in MIL-B-81705 type I,II, or III material, if required (see 6.2.2)	See 3.1.5
Non-ESD sensitive resistors			
Axial lead, film composition (tape and reel) MIL-R-10509 MIL-R-22684 MIL-R-39017 MIL-R-49462	No requirement	No requirement	See 3.1.5
(bulk)	No requirement	No requirement	See 3.1.5
Axial lead, composition MIL-R-11 MIL-R-39008	No requirement	No requirement	See 3.1.5

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Table III. Shipments to non-Government facilities - Continued.

Type	Physical protection	Bag/material construction	Marking
Non-ESD Sensitive Resistors - Continued			
Axial lead, fixed and varaible (including lug types) MIL-R-26 MIL-R-93 MIL-R-18546 MIL-R-39005 MIL-R-39007 MIL-R-39009 MIL-R-49465	No requirement	No requirement	See 3.1.5
Temperature sensitive MIL-T-23648	No requirement	No requirement	See 3.1.5
Variable, trim pots, excluding those with antifriction bearings MIL-R-22097 MIL-R-27208 MIL-R-39015 MIL-R-39035	No requirement	No requirement	See 3.1.5
Variable with antifriction bearings	MIL-P-116 method IIC, cushioning in accordance with 3.1.1.4, MIL-D-3464 desiccant, MS-20003 humidity indicator	No requirement	See 3.1.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Inspection conditions. All inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-202.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used are in accordance with the applicable requirements specified herein.

4.4 First article inspection. When specified (see 6.2), first article inspection shall be performed by the contractor, after award of contract and prior to production, at a time and location acceptable to the Government. First article inspection shall not be required:

a. When there have been no changes in materials, processes, or packaging design that will adversely affect item protection since the last recorded inspection.

b. When detailed packaging instructions are furnished by the acquisition activity.

c. When level C protection is specified.

d. When a prior successful inspection was conducted on a like item and pack (subject to the approval of the administrative contracting officer).

4.4.1. Sample size. One sample unit consisting of a level A or level B, as applicable, fully packed shipping container shall be submitted for first article inspection. The sample for the rough handling test shall consist of the pack selected for first article inspection. The sample for the leakage test shall be five unit packs selected at random from the first article exterior pack (shipping container).

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in tables IV and V. The leakage test, when applicable, shall follow the rough handling test.

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4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.4.4 Resubmission of first article sample. If the sample fails to pass first article inspection, the contractor shall change the preservation and packing processes to correct the cause of the deficiency. First article inspection shall be performed on a corrected sample to prove that the corrective action is acceptable.

Table IV. Visual and dimensional inspections.

Major defects	Requirement paragraph	Method paragraph
Preservation and packing materials not conforming to referenced specification requirements	3.1.1, 3.1.2, 3.1.3, and 3.2	4.6.1
Improper field force protection (shielding)	3.1.8 and 3.2	
Punctured or improperly fabricated barrier bag	3.1.1.4 and 3.2	
Uncleaned or improperly cleaned items	3.1.2.1.1 and 3.2	
Incorrect preservation method	3.1.2.1.3 and 3.2	
Wrong quantity per unit pack	3.1.2.1.4	
Nonusable or incorrect kind or application of intermediate containers	3.1.2.1.7 and 3.2	
Improper box closures	3.1.3 and 3.2	
Omitted, incorrect, or illegible marking	3.1.5 and 3.2	

TABLE V. Functional inspections.

Test	Requirement paragraph	Method paragraph
Rough handling (when specified)	3.1.6.1.1	4.6.2.1
Leakage (when applicable)	3.1.6.1.2	4.6.2.2

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4.5 Quality conformance inspection. This inspection shall consist of the inspections and tests specified in tables IV and V, respectively.

4.5.1 Shipments to Government activities.

4.5.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of unit or exterior (shipping) packs produced under essentially the same conditions and offered for inspection at one time. For the purpose of selecting samples to be inspected and tested for compliance with the requirements of this specification, either items in process or completed packs except as stated herein, may be combined into lots without regard to individual items, contracts, or the quantities therein. Unit packs of the same size and made from the same packaging materials may be grouped together except when item complexity, item value, or the complexity of the preservation method warrants that the inspection of such items be performed on a separate basis. A separate application of the sampling or inspection procedure shall be made on these items. The combination of items to be subjected to inspection shall be determined by either the Government or the contractor, subject to the approval of the Government.

4.5.1.2 Visual and dimensional inspection. Visual and dimensional inspection shall consist of those inspections specified in table IV.

4.5.1.3 Sample plan. A sample of five packages shall be randomly selected. If one or more defects are found, the inspection lot shall be screened and the defective packages removed. A new sample of five packages shall be randomly selected. If any defects are found in the second sample, the inspection lot shall be rejected and shall not be supplied to the resistor specification.

4.5.1.4 Disposition of sample units. Sample units which have passed all the inspections specified in table IV may be delivered on the contract, provided the lot is accepted.

4.5.2 Shipments to non-Government facilities. Shipments to non-Government facilities shall be for the visual and dimensional inspections specified in table IV.

4.5.3 Functional inspection. Functional inspection shall consist of the tests specified in table V.

4.5.3.1 Sampling plan. Sampling plans shall be as follows:

a. One sample unit for the rough handling test shall be selected whenever the design of the item or packaging is changed.

b. For unit packs requiring waterproof or water-vaporproof barriers, five sample units for the leakage test shall be selected daily, at random, from the first lot processed each day. Five additional samples shall be selected at random from the day's total production.

c. The leakage test shall also be performed following the rough handling test on unit packs requiring waterproof or water-vaporproof barriers. Five sample units or the number of units contained within the shipping container (if less than five) shall be selected.

4.5.3.2 Failures. One or more failures shall be cause for rejection of the lot.

4.5.3.3 Disposition of sample units. Sample units which have passed the inspections specified in table V may be delivered on the contract, if the lot is accepted and opened packs have been reprocessed.

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4.5.3.4 Noncompliance. If a sample fails to pass the inspections specified in table IV, the contractor shall take corrective action on the materials or processes or both, as warranted, on all units, intermediate, and exterior (shipping) packs which can be corrected and which were processed under essentially the same conditions, with essentially the same materials, and which are considered subject to the same failure. Acceptance of the unit, intermediate, and exterior packs shall be discontinued until corrective action has been taken and the applicable inspections specified in table IV have been repeated on additional sample units (all inspections or the inspection which the original sample failed, at the option of the Government). Inspections specified in table IV may be reinstituted; however, final acceptance shall be withheld until the reinspection in accordance with table IV has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the administrative contracting officer.

4.6 Methods of inspections and tests.

4.6.1 Visual and dimensional inspections. Unit, intermediate, and exterior packs shall be examined to verify that the materials, designs, methods, physical limitations, marking, and workmanship are in accordance with the applicable requirements (see 3.1.1 through 3.1.8).

4.6.2 Functional tests.

4.6.2.1 Rough handling. Packs shall be subjected to the applicable rough handling tests and the interpretation of results or cause for rejection as specified in 3.1.6.1.1 and MIL-P-116.

4.6.2.2 Leakage. When a waterproof or water-vaporproof barrier is required, the unit pack shall be subjected to the applicable leakage test and interpretation of results as specified in 3.1.6.1.2 and MIL-P-116.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The preservation, packing, and marking specified herein are intended for direct shipments to the Government and to OEM and packaging distributors. Unless otherwise designated, the general requirements (3.1) and marking requirements (3.1.5) are applicable for the preparation of military specification on resistors and accessories for shipment from the parts manufacturer to non-Government facilities.

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6.2 Acquisition requirements.

6.2.1 Shipments to Government activities. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation and packing (see 3.1.2 and 3.1.3).

c. Quantity per unit pack, if other than specified (see 3.1.2.1.4).

d. If a unitized load is not required for shipment to one destination when total quantities are equal to 40 cubic feet or more (see 3.1.4).

e. Whether any other standard or special marking is required (see 3.1.5).

f. If FED-STD-123 is required for civil agency marking (see 3.1.5.1).

g. If a rough handling test is required (see 3.1.6.1.1).

h. If resistor functional tests are required (see 3.1.6.1).

i. If the contractor is not responsible for the performance of all inspection requirements (see 4.1).

j. If first article inspection is not required (see 4.4).

6.2.2 Shipments to non-Government activities. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation and packing and specify the packaging material to be used if the material is sensitive to ESD charges (see MIL-E-17555) (see 3.2.2.2).

c. Whether any standard or special markings are required.

d. Quantity per unit pack.

6.3 Inspection for first article. The sample pack submitted for first article inspection (when satisfactorily performed as specified in 4.4) will serve as the production standard for subsequent packaging operations. The contractor should inform the acquisition activity or the activity administering the contract of the time and location of the inspection so that the Government representative will have an opportunity to witness the tests.

6.4 Definition. A hermetic seal is defined as a fusion of metal to metal, glass to glass, or metal to glass.

6.5 Conditions for use of level B preservation. When level B preservation is specified (see 3.1.2), this level of protection should be reserved for the acquisition of resistors and accessories for resupply worldwide under known favorable handling, transportation, and storage conditions (see 3.1.2.2).

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6.6 Subject term (key word) listing.

Electrostatic discharge protection
Environmental field force protection
Film resistors
Fixed resistors
Hermetically sealed resistors
Nonhermetically sealed resistors
Packing
Physical protection
Preservation
Resistor accessories
Resistor chips
Resistor networks
Shielding
Special marking
Variable resistors

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
DLA - ES

Review activities:
Army - AR, CR, MI, SM
Navy - SH
Air Force - 99

User activities:
Army - AT, AV, ME
Air Force - 17

Preparing activity
Army - ER

Agent
DLA - ES

(Project PACK-0934)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-39032D	91/02/11

3. DOCUMENT TITLE
Resistors, Packaging of

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
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